Open in app ↗

Medium Search Write

Member-only story

🏀 Analyzing NBA Data Using Python and APIs

Mastering Data Analysis through Basketball Data

Ben Ballard 3 min read · Feb 19, 2023

132 2

If you're a fan of basketball, you probably love watching NBA games and following your favorite players. But did you know that you can also analyze NBA data using Python and a powerful API? In this blog post, I'll show you how to use the NBA_API to access NBA data, perform statistical analysis, and create visualizations.

The NBA has become a staple in American culture and as technology has progressed, accessing NBA data has become increasingly easier. There are several NBA APIs available on the web, but for the purpose of this analysis, we will focus on the nba_api package. The `nba_api` is an API Client for `www.nba.com`. This package stated goal is to make the APIs of NBA.com easily accessible and provide accessible documentation. This package is open-source.

NBA API

Using the nba_api package, it is simple to write a script or program that can request and parse NBA data. The nba_api provides many methods such as get_players(), playercareerstats(), etc. These methods can be used to pull data from the NBA API and transform it into a pandas dataframe.

Let's take a look at an example. The following script pulls data for the top 500 scorers by PTS column, groups them by player_name and Player_ID columns, and calculates their averages for MIN, FGM, FGA, FTM, FTA, PTS, FG3M, FG3A, and GP columns.

```
from nba_api.stats.endpoints import leagueleaders
import pandas as pd

try:
    # Pull data for the top 500 scorers
    top_500 = leagueleaders.LeagueLeaders(
        season='2023-24',
        season_type_all_star='Regular Season',
        stat_category_abbreviation='PTS'
    ).get_data_frames()[0][:500]

    # Correct column names for grouping
    avg_stats_columns = ['MIN', 'FGM', 'FGA', 'FTM', 'FTA', 'PTS', 'FG3M', 'FG3A
    top_500_avg = top_500.groupby(['PLAYER', 'PLAYER_ID'])[avg_stats_columns].me

    # Inspect the first few rows of the averaged stats
    print(top_500_avg.head())

except Exception as e:
    print(f"An error occurred: {e}")
```

The `leagueleaders.LeagueLeaders()` method takes in parameters such as the season and stat category abbreviation to pull data for the top 500 players by points scored per game for the specified season. The `get_data_frames()`

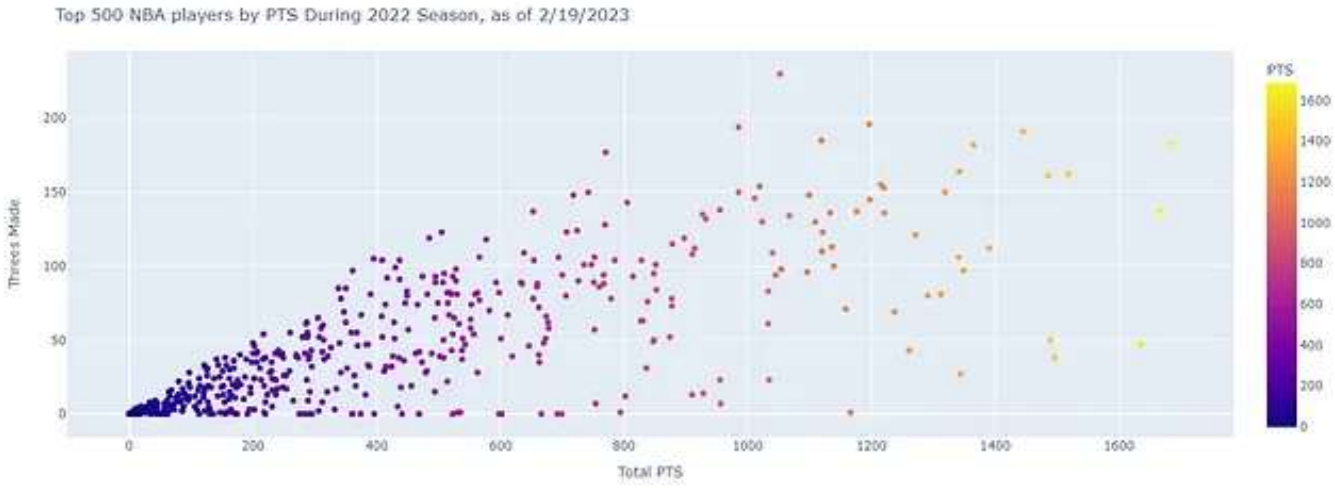
method returns a list of data frames, with the first item in the list containing the data we want. We then filter out the top 500 players and group them by name and player ID using the `groupby()` method. Finally, we calculate the average values for the desired columns.

Once we have our data in a pandas dataframe, we can perform additional analysis on it. For example, we can plot the total points scored versus the number of three-pointers made using plotly.

```
import plotly.express as px

# Reset index to turn 'PLAYER' and 'PLAYER_ID' back into regular columns
df_for_plotting = top_500_avg.reset_index()

# Create a scatter plot with colors based on 'PTS'
fig = px.scatter(
    df_for_plotting,
    x='PTS',
    y='FG3M',
    hover_name='PLAYER',
    color='PTS',  # This line makes the color of the points depend on the 'PTS'
    color_continuous_scale=px.colors.sequential.Viridis  # Optional: choose a co
)

fig.show()
```

Visualization

This will display an interactive scatter plot of the average number of three-pointers made versus the total points scored for each player in the top 500.



Top 500 NBA Player PTS scored for 2023–24

The scatter plot shows a scatter plot of Total PTS, at this point in the season 2/19/2023, and total threes made. To get a better sense of the top 5 players in scoring, I've included a table below, as well as the python line that will get you the below.

```
# Assuming top_500_avg is your DataFrame with average stats
top_10 = top_500_avg.sort_values(by='PTS', ascending=False).head(10)
top_10
```

PLAYER	PLAYER_ID	MIN	FGM	FGA	FTM	FTA	PTS	FG3M	FG3A
Luka Doncic	1629029	1251.0	388.0	800.0	231.0	297.0	1142.0	135.0	355.0
Giannis Antetokounmpo	203507	1268.0	420.0	692.0	277.0	410.0	1132.0	15.0	63.0
Shai Gilgeous-Alexander	1628983	1218.0	395.0	716.0	266.0	300.0	1099.0	43.0	122.0
Nikola Jokic	203999	1266.0	374.0	652.0	181.0	221.0	969.0	40.0	115.0
Jayson Tatum	1628369	1245.0	321.0	669.0	198.0	243.0	951.0	111.0	297.0
Jalen Brunson	1628973	1317.0	334.0	712.0	177.0	213.0	949.0	104.0	244.0
Trae Young	1629027	1246.0	292.0	681.0	248.0	290.0	946.0	114.0	315.0
Stephen Curry	201939	1161.0	294.0	658.0	189.0	205.0	935.0	158.0	397.0
Joel Embiid	203954	922.0	314.0	587.0	276.0	313.0	935.0	31.0	86.0
Anthony Davis	203076	1302.0	349.0	621.0	212.0	262.0	924.0	14.0	39.0

2023–24 NBA Points Leaderboard

In conclusion, the **nba_api** package is a great tool for accessing and analyzing NBA data. With its many methods, it's a great way to grab fun data to conduct your analysis. With powerful tools such as pandas and plotly, it is easy to perform gain insights and product visualizations. Whether you are a casual NBA fan or a seasoned analyst, the nba_api package is worth checking out.

If you want to go deeper, I've created a more detailed post on the NBA API and manipulating the data in python. This post provides more python code, data pulls, and visualizations for NBA data. Thanks for reading!

How to Analyze NBA Stats with the NBA API and Python

medium.com

Data Science | Basketball | Data Analytics | Python | API

Written by Ben Ballard

203 followers · 365 following

Here for Data Science and Machine Learning. MS Data Science @UVA | Boomer | Mavs | Working on NBAanalytics.com.

Edit profile

Responses (2)

Ben Ballard

What are your thoughts?

Harold Uribe Romero
Jan 17, 2024

Hi Ben, I'd been searching for NBA data in order to try to enter in this data field of sports analytics. Do you know some resources to learn about the sports analytics used in NBA? thanks in advance

2 1 reply Reply

Wynne Philpott
Feb 22, 2024

Hi Ben - If i want to track Mavericks wins, which endpoint would i use? I counted 138 sunmodules under nba_api.stats.endpoints so am unsure where to start.

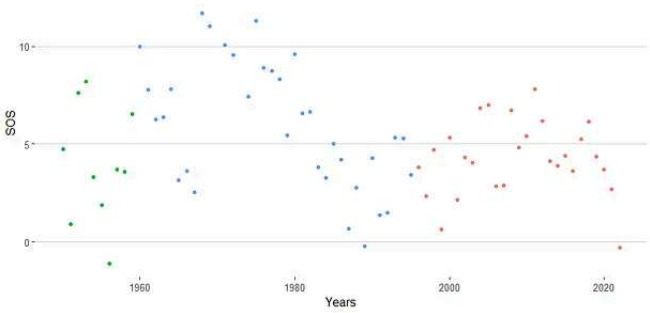
1 1 reply Reply

More from Ben Ballard

Ben Ballard

Lebron James Career-Year from Beyond the Arc

Dec 8, 2023 4



In Data Science Cowboy by Ben Ballard

OU Strength of Schedule Over Time

For my next analysis, I took a look at the strength of schedule variable. I used the R...

Sep 17, 2022

Ben Ballard

How to Analyze NBA Stats with the NBA API and Python

Ben Ballard

Unlocking Sports Betting with Python: The Odds API

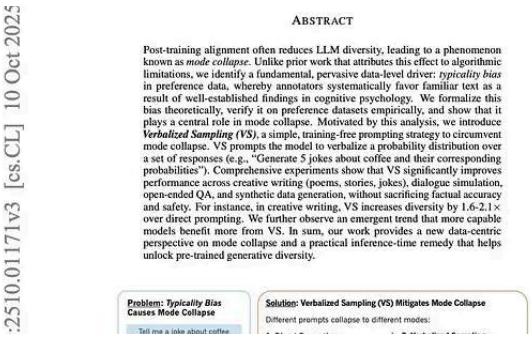
Harnessing Live Data for Smarter Wagering

Dec 11, 2023 92 1

Dec 16, 2023 96

See all from Ben Ballard

Recommended from Medium



In Generative AI by Adham Khaled

Stanford Just Killed Prompt Engineering With 8 Words (And I...

ChatGPT keeps giving you the same boring response? This new technique unlocks 2×...

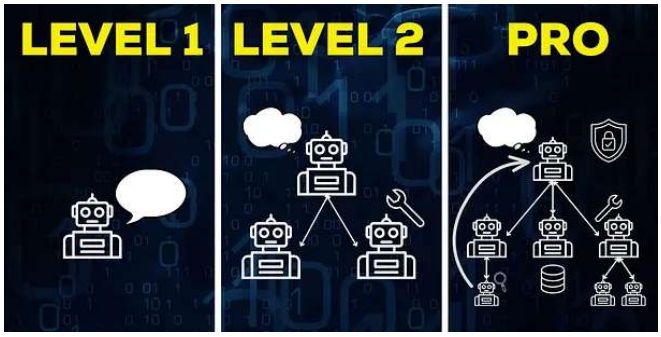
Oct 19, 2025 23K 616

In AWS in Plain English by Khushbu Shah

Data Engineering Design Patterns You Must Learn in 2026

These are the 8 data engineering design patterns every modern data stack is built on....

Jan 5 898 19

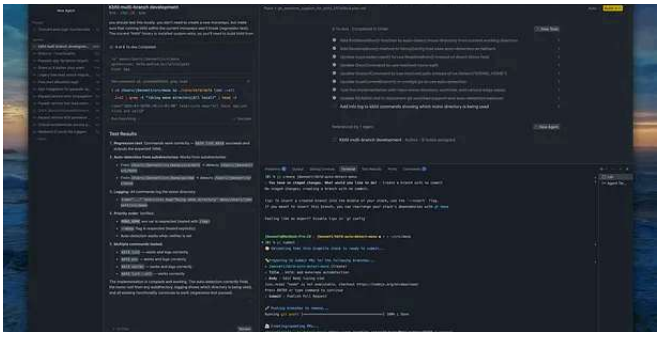


In Data Science Collective by Marina Wyss

AI Agents: Complete Course

From beginner to intermediate to production.

Dec 6, 2025 3.4K 116



Jacob Bennett

The 5 paid subscriptions I actually use in 2026 as a Staff Software...

Tools I use that are (usually) cheaper than Netflix

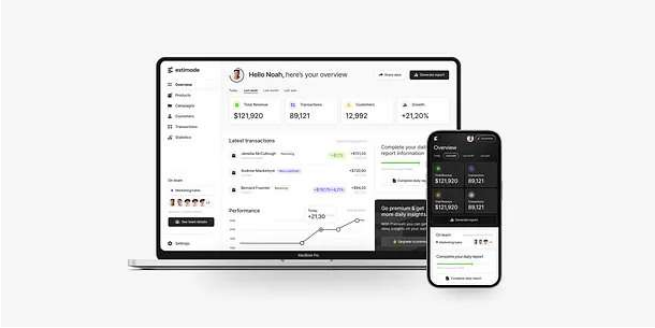
Jan 18 2.6K 68

Will Lockett

The AI Bubble Is About To Burst, But The Next Bubble Is Already...

Techbros are preparing their latest bandwagon.

Sep 14, 2025 22K 946



Michal Malewicz

The End of Dashboards and Design Systems

Design is becoming quietly human again.

Nov 26, 2025 5.5K 212

See more recommendations